An age-structured contact model of sick-leave and school-absence effects on seasonal influenza in Korea

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Contents

Section 1

Background

The evidence for sick leave is limited

- Keeping infectious individuals out of a shared setting is a standard non-pharmaceutical response to influenza:
  - **school exclusion** for symptomatic children;
  - **sick leave** for symptomatic workers.
- Keeping children out of school is comparatively well-studied. Keeping adults home from work is **far less established**.
- The intuitive case for both is the same: an infectious person who stays home cannot infect the colleagues or classmates they would have met — so evaluations have concentrated on the **size and direction of that reduction**.
- Here we evaluate both policies with a single, common method, in one model and for the same seasons.

Key modelling assumptions

For an age-structured respiratory pathogen, a single aggregate effect size leaves out four things:

- **Heterogeneity.** Susceptibility, attack rate, and contacts vary sharply by age; children are among the most susceptible and, in most seasons, the most infectious group.
- **Redistribution.** Sending an infectious person home does not end their transmission — it **relocates** it into the household, whose age mix differs from the setting they left.
- **Presymptomatic transmission.** A share of influenza A spreads before symptoms, beyond the reach of any symptom-triggered policy.
- **Timing.** The household is always open, even when schools are not, so a policy that shifts transmission home behaves differently in term than in the winter break.

Aims

We built an age-structured SVEIR model of seasonal influenza in the Seoul Capital Area to move from "how much is averted" to "**whom** does the policy move risk toward, **with what certainty**, and **when**." The model lets us:

(i) compare sick-leave and school-absence policies at matched intensity, within a single consistent methodology, with credible intervals;

(ii) identify the conditions — how much transmission occurs at work, and how crowded the home becomes — under which sick leave is net-beneficial versus net-harmful;

(iii) decompose outcomes by age and by metric (rate vs. count);

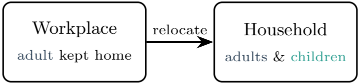
(iv) ask how the same sick-leave policy behaves in different phases of the epidemic, split by the winter-break calendar.

Section 2

Model assumptions

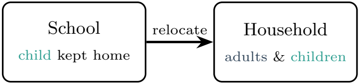
Household spillover: the mechanism

Sick leave



work transmission removed (adults ↓); added home transmission, amplified by children's higher susceptibility, raises child infection

School absence



school transmission removed (children ↓↓); added home transmission reaches co-resident adults

- Home contacts: a child's are ≈80% with adults; an adult's are ≈78% with other adults.
- The net direction differs because the removed source and the household age mix differ.

Household spillover: formulation

Baseline attendance is a retention fraction p (kept attending while symptomatic); the policy lowers p, sending a fraction $1 - p$ home. Only the symptomatic block I_{23} is affected; the home-channel symptomatic prevalence is scaled by

$$1 + \kappa_b \, (1 - p)_b, \qquad (1 - p)_b = \begin{cases} 1 - p_{\text{school}}, & \text{students}, \\ \rho_b \, (1 - p_{\text{work}}), & \text{workers}, \\ 0, & 70+, \end{cases}$$

- κ_b = per-group increase in effective household transmission from a symptomatic person staying home; ρ_b = employment rate (students $\rho_b = 0$, no commute).
- κ is **derived, not fitted** — from the 2024 Korean Time-Use Survey (increase in home-stay time on a non-attendance vs. attendance day). Spillover is an input, so the result is not circular.

Presymptomatic transmission

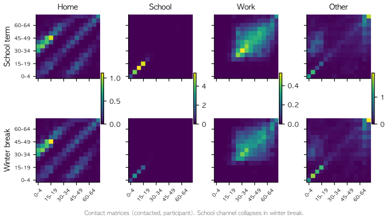
- The Erlang(3) infectious period $I_1 \to I_2 \to I_3$ gives a distinct **presymptomatic** sub-stage I_1: infectious but not yet symptomatic.
- Symptoms — and any symptom-triggered behaviour (care-seeking, absenteeism) — begin at $I_1 \to I_2$. Both policies act only on I_2, I_3; the observation model (NHIS) is placed at symptom onset.
- I_1 transmits at reduced weight $w = 0.22$ ($\Rightarrow$ presymptomatic share $w/(w+2) \approx 9.9\%$), following a recent influenza A household study.
- **Consequence:** presymptomatic transmission sets a **ceiling** on what any symptom-based policy can achieve — much of a sick worker's transmission has already occurred, or occurs through I_1 contacts the policy never reaches.

The school calendar acts on contacts, not on β

- Each channel's matrix is a trapezoidal-in-time blend of a school-term and a winter-break matrix:

  $$C^c_{ab}(t) = (1-h(t))\, C^{c,\mathrm{term}}_{ab} + h(t)\, C^{c,\mathrm{vac}}_{ab},$$

  with $h(t)$ the winter-break ramp.
- Matrices are from a nationwide Korean (NIMS) contact survey, term and break measured separately.
- Putting the calendar in $C(t)$ (not β_{school}) keeps the baseline home spillover $\equiv 1.0$ across the year — contact frequency and intervention intensity never contaminate each other.



Contact matrices by channel, school term vs. winter break.

Susceptibility, seasonal forcing, vaccination

- **Age susceptibility ϕ_a:** linear U-shape — child anchor 2.0 (0–4), working-age floor 1.0 (20–44), elderly 1.5 (70+), intermediate bands interpolated.
- **Seasonal forcing:** $s_f(t) = 1 + 0.9\cos\left(\frac{2\pi(t-105)}{365}\right)$, peaking in mid-December.
- **Vaccination:** time-distributed leaky-protection hazard $v_a(t) = \frac{-\ln(1-C_a)}{Z}\,\mathrm{density}(t)$, reaching an age-specific target coverage C_a (0.30–0.82; children and elderly higher); efficacy $\mathrm{VE} = 0.5$.

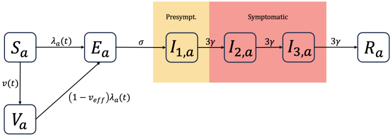
Section 3

Model and methods

Age-structured SVEIR compartmental structure

For each of 15 five-year age groups a:

$$\frac{dS_a}{dt} = -\lambda_a S_a - v(t) S_a,$$
$$\frac{dV_a}{dt} = v(t) S_a - (1 - \mathrm{VE}) \lambda_a V_a,$$
$$\frac{dE_a}{dt} = \lambda_a S_a + (1 - \mathrm{VE}) \lambda_a V_a - \sigma E_a,$$
$$\frac{dI_{1,a}}{dt} = \sigma E_a - 3\gamma I_{1,a},$$
$$\frac{dI_{2,a}}{dt} = 3\gamma I_{1,a} - 3\gamma I_{2,a},$$
$$\frac{dI_{3,a}}{dt} = 3\gamma I_{2,a} - 3\gamma I_{3,a},$$
$$\frac{dR_a}{dt} = 3\gamma I_{3,a}.$$



Latent period $1/\sigma = 2$ d; mean infectious period $1/\gamma = 4$ d over an Erlang(3) chain. I_1 presymptomatic (weight w), I_2, I_3 symptomatic (policy acts here).

Force of infection: four contact channels

With $I_{23,b} = I_{2,b} + I_{3,b}$, the channel hazards are

$$\begin{aligned}
\lambda_a^{\text{work}} &= \phi_a s_f(t)\,\beta_{\text{work}} \textstyle\sum_b C_{ab}^{\text{work}}(t)\,\frac{wI_{1,b}+p_{\text{work}}I_{23,b}}{N_b},\\
\lambda_a^{\text{school}} &= \phi_a s_f(t)\,\beta_{\text{school}} \textstyle\sum_b C_{ab}^{\text{school}}(t)\,\frac{wI_{1,b}+p_{\text{school}}I_{23,b}}{N_b},\\
\lambda_a^{\text{home}} &= \phi_a s_f(t)\,\beta_{\text{home}} \textstyle\sum_b C_{ab}^{\text{home}}(t)\,\frac{wI_{1,b}+(1+\kappa_b(1-p)_b)I_{23,b}}{N_b},\\
\lambda_a^{\text{other}} &= \phi_a s_f(t)\,\beta_{\text{other}} \textstyle\sum_b C_{ab}^{\text{other}}(t)\,\frac{wI_{1,b}+I_{23,b}}{N_b},
\end{aligned}$$

with $\lambda_a = \sum_c \lambda_a^c$. The presymptomatic block wI_1 passes through every channel unaffected; policy retention $p_{\text{work}}, p_{\text{school}}$ and the home crowding term act only on I_{23}.

Parameters: fixed constants

Symbol	Description	Value	Note
$1/\sigma$	Latent period	2.0 d	standard
$1/\gamma$	Mean infectious period	4.0 d	standard
n	Erlang sub-stages	3	rate 3γ
w	Presymptomatic infectiousness	0.22	influenza A presymptomatic share 9.6%
VE	Vaccine efficacy	0.5	standard
$s_f(t)$	Seasonal forcing	$1 + 0.9\cos\frac{2\pi(t-105)}{365}$	peak mid-December
C_a	Target vaccine coverage	age-specific, 0.30–0.82	children & elderly higher

The infectious period is an Erlang(3) chain $E \to I_1 \to I_2 \to I_3 \to R$; I_1 is presymptomatic (weight w, not subject to policy), I_2, I_3 are symptomatic (policy acts here).

Parameters: age-specific vectors

Age band	ϕ_a susceptibility	$\gamma_{\text{report},a}$ reporting	$R(0)_a$ initial immune	κ_a crowding	ρ_a employment
0–4	2.00	0.40	0.10	0.34	0
5–9	1.75	0.40	0.10	0.34	0
10–14	1.50	0.25	0.10	0.34	0
15–19	1.25	0.18	0.10	0.34	0
20–44	1.00	0.18	0.40	0.40	measured, $\to\approx 0.74$
45–49	1.08	0.18	0.60	0.40	measured
50–64	1.17–1.33	0.18	0.60	0.40	measured
65–69	1.42	0.25	0.65	0.40	measured
70+	1.50	0.25	0.65	0	0

ϕ_a: three literature anchors interpolated. κ_a: student 0.34, adult 0.40, $\kappa_{70+} = 0$; derived from the Korean Time-Use Survey. ρ_a: employment rate (Statistics Korea); students take $\rho_a = 0$, so κ acts on the home channel only.

Data and seasons

- **Incidence:** NHIS influenza claims for the Seoul Capital Area, via the NHISS research platform; aligned to symptom onset.
- **Seasons:** three "normal" single-wave seasons — 2016–17, 2017–18, 2019–20. Excluded: 2015–16 (atypical onset), 2018–19 / 2022–23 (double-peaked).
- **Contacts:** age-structured school-term and winter-break matrices from the NIMS contact survey.
- **Populations:** season-specific resident registration (Statistics Korea, KOSIS; June of each peak year), summed over Seoul, Incheon, and Gyeonggi into one aggregated region.

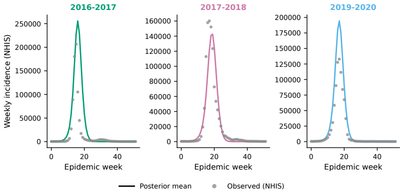
Identifiability and policy set-up

- **Workplace transmission is not identifiable from one season:** freely estimated, β_{work} rails toward zero in 88–94% of single-season fits (home and work matrices are similar among working-age adults). The multi-season joint fit closes this gap.
- **How a policy is applied in the model:** during a set time window, a fraction of symptomatic people who would have attended work (sick leave) or school (school absence) instead stay home; the two operators act on separate channels and only on the symptomatic stages I_{23}.
- Comparisons use posterior forward simulation, $N = 500$ draws per season, sweeping intensity $p \in \{1, 0.8, 0.6, 0.4, 0.2\}$ and the adult crowding coefficient κ from 0.2 to 1.0.
- The epidemic splits by the winter-break calendar into a school-term and a winter-break window; we report window-length-robust quantities.

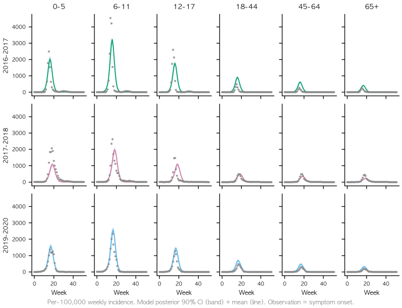
Section 4

Results

The calibrated model reproduces all three seasons

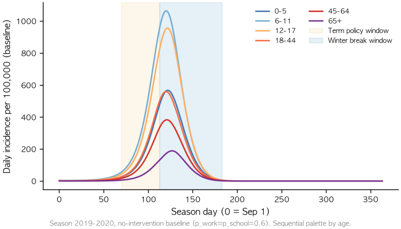


Aggregate weekly incidence (posterior mean vs. observed).

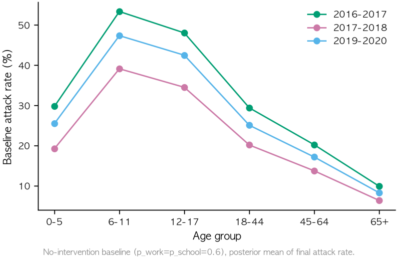


By-age fit; over-prediction of epidemic width is most visible in the sharp 2016–17 season and the youngest band.

Baseline: most infection occurs in school-age children



Modelled age-specific epidemic curves at baseline.



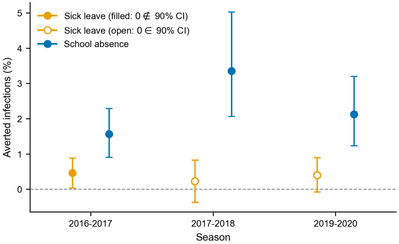
Baseline final attack rate by age.

School-age children carry the highest attack rates and peak early; the school channel drives the epidemic despite a small fitted share ($\pi_{\text{school}} \approx 0.065$).

School absence is effective; sick leave is small and uncertain

Season	Sick leave	School absence
16–17	$+0.46^{\star}$	$+1.56^{\star}$
17–18	$+0.23$	$+3.36^{\star}$
19–20	$+0.40$	$+2.12^{\star}$

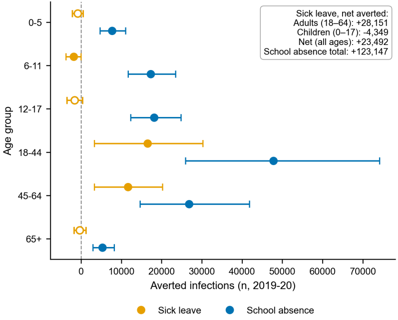
Averted infection (%). $\star$: $0 \notin 90\%$ CI.



- School absence excludes zero in every season; sick leave's CI **includes zero in 2 of 3** seasons.
- Sick leave is not shown to be harmful (point estimates positive), but its population benefit is small and not statistically established.

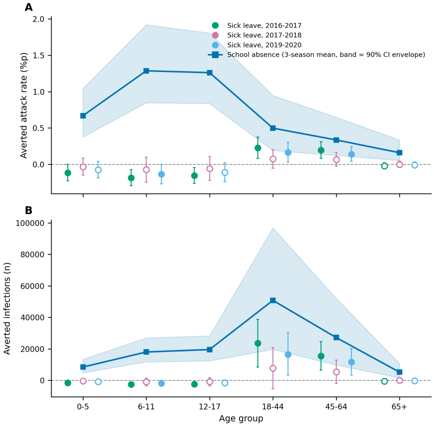
School absence is also the larger lever

- In absolute counts (2019–20): school absence averted ≈123,147 infections vs. sick leave's net ≈23,492 — roughly **fivefold**.
- The school channel carries a small transmission share but is child-dominated, so suppressing it yields disproportionate returns.
- Because sick leave's effect sits near zero, we report the direction rather than a fixed multiple.



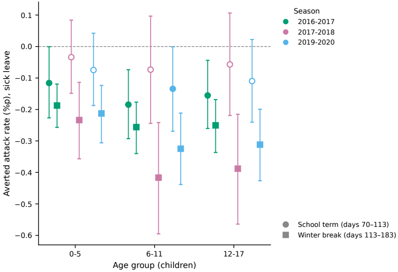
Sick leave redistributes from adults toward children

- **Per-capita (top, averted attack rate):** adults down, children **below zero** (infection added). Same direction in **3 of 3** seasons; the elderly unaffected.
- **Absolute counts (bottom):** adult bands dominate the averted total (18–44 ∼10 million; 2019–20: ≈28k adult averted vs. ≈4k added child).
- "Who benefits" thus depends on the metric; the effect is small and only partly significant — a tendency, not established harm.



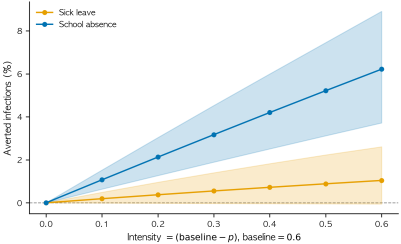
The transfer toward children is larger during the winter break

- Sick-leave spillover flows through the home channel, which the winter break does not close.
- Children are home more during the break, so the relocated transmission concentrates on them — a larger child-ward transfer than during the school term.
- Direction robust (positive in **749/750** sensitivity draws); magnitude scales with household crowding.



Dose–response: school absence dominates at every intensity

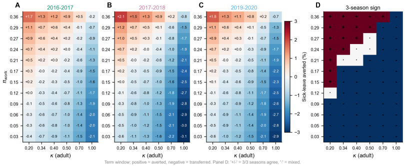
- Total averted infection rises with policy intensity for both levers.
- School absence dominates across the full intensity range at the identified crowding scale.



Season 2019-2020, term window. Band = 90% CI from N=30 posterior samples.

Sensitivity: crowding κ and workplace share

- Sick leave turns net-harmful only above $\kappa \approx 0.51$.
- Our time-use–based estimate ($\kappa \approx 0.34$–0.40) sits **below** this threshold, so the regime is net-beneficial, though small.
- But larger household-contact increases have been reported (+50 to +100%; Ferguson et al.), which would reach or exceed $\kappa \approx 0.51$ — a net-harmful regime cannot be ruled out under those assumptions.



Section 5

Discussion

School absence works; sick leave is uncertain

- School-based measures are more effective and more certain across all seasons and age groups, and carry no household-spillover hazard.
- Sick leave's population benefit is small and not robustly established. This follows from **household spillover**: much of the transmission removed at work is re-created at home.
- Its benefit is the fragile difference between transmission removed at work and transmission re-created at home.

Redistribution, metric, and timing

- **Redistribution.** Sick leave moves risk from adults toward school-age children (direction robust in 3 of 3 seasons; magnitude small and only partly significant).
- **Metric.** In per-capita terms children bear it; in absolute counts adults dominate the benefit. Both readings are correct — the choice of ruler is itself a policy judgement.
- **Timing.** The child-ward transfer intensifies during the winter break, when the home channel stays open and children are home more.

Policy implications and reframing

- **Prioritize school-based measures** where feasible: more effective, more certain, no spillover hazard.
- **Where sick leave is used,** pair it with household-isolation support, especially during school holidays when its child-ward transfer is largest.
- **Reframe evaluation:** appraise absenteeism policy by **whom** it moves risk toward, **with what certainty**, and **when** — not by an aggregate effect size alone.

Limitations

- Presymptomatic weight taken from influenza A; influenza B shows no clear presymptomatic transmission (an A-based weight is conservative).
- Crowding κ derived from time-use data, not a contact diary of sick individuals; both κ and the baseline retention p are swept in sensitivity.
- The contact survey was conducted near the Lunar New Year holiday (weekdays close to the holiday), so the contact matrices may carry a holiday-related bias.
- Absence is modelled over the entire infectious period, which may differ from actual absence durations.
- Sick leave's near-zero effect makes its exact magnitude and count ratios imprecise — we emphasize direction and certainty.

Thank you.

Q&A